

2.6 Motion Along a Curve

Motivation: Suppose that a particle moves through 3 space so that its position vector at time t is $\mathbf{r}(t)$. As the particle moves it has a velocity in the x , y , and z direction. How should we go about calculating the velocity in each direction?

The answer to this question is that we should find $\mathbf{r}'(t)$. Each component function of this vector function will give us the velocity in the x , y , and z direction, respectively. This basic idea leads to the following definitions.

Definition: Suppose a particle moves through 3 space so that its position at time t is given by the twice differentiable vector function $\mathbf{r}(t)$.

- The velocity vector of the particle at time t is $\mathbf{v}(t)$ and $\mathbf{v}(t) = \mathbf{r}'(t)$.
- The acceleration vector of the particle at time t is $\mathbf{a}(t)$ and $\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t)$.
- The speed of the particle at time t is defined to be $\|\mathbf{v}(t)\|$.

Remark: One should notice that $\|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\| = \frac{ds}{dt}$. So, we see that speed as it is defined above is actually the rate of change of distance with respect to time which is a typical way in which speed is defined.

Example 1: Suppose that a particle starts at an initial position of $\langle 1, 1, 1 \rangle$ with an initial velocity of $\mathbf{v}(0) = 6\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$. Also suppose that the acceleration vector of the particle is $\mathbf{a}(t) = 4t\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ where $t \geq 0$.

a) Find the velocity vector for the particle at time t .

We note that $\mathbf{v}'(t) = \mathbf{a}(t)$. This means that

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt = \left(\int (4t) dt \right) \mathbf{i} + \left(\int 3 dt \right) \mathbf{j} + \left(\int (-1) dt \right) \mathbf{k} = (2t^2) \mathbf{i} + (3t) \mathbf{j} - (t) \mathbf{k} + \mathbf{C} \text{ where } \mathbf{C} \text{ is a constant vector.}$$

Now, we must use the fact that $\mathbf{v}(0) = 6\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ to find $\mathbf{C}$.

We note that $\mathbf{v}(t) = (2t^2) \mathbf{i} + (3t) \mathbf{j} - (t) \mathbf{k} + \mathbf{C}$ implies $\mathbf{v}(0) = \mathbf{C}$.

Thus, $\mathbf{C} = 6\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, and $\mathbf{v}(t) = (2t^2 + 6) \mathbf{i} + (3t + 2) \mathbf{j} + (2 - t) \mathbf{k}$.

b) Find the position vector for the particle at time t .

We note that $\mathbf{r}'(t) = \mathbf{v}(t)$. This means:

$$\begin{aligned} \mathbf{r}(t) &= \int \mathbf{v}(t) dt = \left(\int (2t^2 + 6) dt \right) \mathbf{i} + \left(\int (3t + 2) dt \right) \mathbf{j} + \left(\int (2 - t) dt \right) \mathbf{k} \\ &= \left(\frac{2}{3} t^3 + 6t \right) \mathbf{i} + \left(\frac{3}{2} t^2 + 2t \right) \mathbf{j} + \left(2t - \frac{1}{2} t^2 \right) \mathbf{k} + \mathbf{C} \end{aligned}$$

where $\mathbf{C}$ is a constant vector. Now, we must use the fact that $\mathbf{r}(0) = \mathbf{i} + \mathbf{j} + \mathbf{k}$ to find $\mathbf{C}$. We note that

$\mathbf{r}(t) = (t^3 + 6t)\mathbf{i} + \left(\frac{3}{2}t^2 + 2t\right)\mathbf{j} + \left(2t - \frac{1}{2}t^2\right)\mathbf{k} + \mathbf{C}$ implies $\mathbf{r}(0) = \mathbf{C}$. Thus, $\mathbf{C} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, and

$$\mathbf{r}(t) = \left(\frac{2}{3}t^3 + 6t + 1\right)\mathbf{i} + \left(\frac{3}{2}t^2 + 2t + 1\right)\mathbf{j} + \left(2t - \frac{1}{2}t^2 + 1\right)\mathbf{k}.$$

c) Find the speed of the particle when $t = 3$.

By definition we know that this is $\|\mathbf{v}(3)\| = \|\langle 24, 11, -1 \rangle\| = \sqrt{24^2 + 11^2 + (-1)^2} = \sqrt{698}$.

Example 2: Suppose that in two dimensions a projectile is fired with an angle of elevation, α , initial velocity $\mathbf{v}_0$, and initial position $\mathbf{r}_0$. Assuming that the only external force is due to gravity, find the position function $\mathbf{r}(t)$ (Note: acceleration due to gravity is 9.8 m/s^2).

The picture to right models the scenario and the parabolic curve represents the trajectory of the particle. We know that the acceleration vector of the particle is $\mathbf{a}(t) = \langle 0, -9.8 \rangle$.

Then, since $\mathbf{v}'(t) = \mathbf{a}(t)$, we have that $\mathbf{v}(t) = \int \mathbf{a}(t) dt = \langle 0, -9.8t \rangle + \mathbf{C}$.

We note that $\mathbf{v}(0) = \mathbf{v}_0$ and $\mathbf{v}(0) = \mathbf{C}$. Thus, $\mathbf{C} = \mathbf{v}_0$.

Now, we would like a formula for $\mathbf{v}(t)$. If $\mathbf{v}_0 = \langle a, b \rangle$, we have the picture at the right.

If $\|\mathbf{v}_0\| = v_0$, simple right triangle trigonometry tells us that $a = v_0 \cos(\alpha)$ and $b = v_0 \sin(\alpha)$.

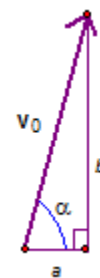
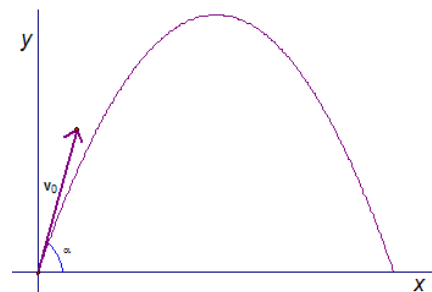
Thus, we have that $\mathbf{v}(t) = \langle 0, -9.8t \rangle + \mathbf{C} = \langle v_0 \cos(\alpha), v_0 \sin(\alpha) - 9.8t \rangle$.

Then, since $\mathbf{r}'(t) = \mathbf{v}(t)$, we know that $\mathbf{r}(t) = \int \mathbf{v}(t) dt = \langle v_0 \cos(\alpha)t, v_0 \sin(\alpha)t - 4.9t^2 \rangle + \mathbf{C}$

where $\mathbf{C} = \mathbf{r}_0$ ($\mathbf{r}_0$ is the initial position of the projectile).

Based on the picture, it appears that the particle starts at the origin.

So, we can say $\mathbf{r}(t) = \langle v_0 \cos(\alpha)t, v_0 \sin(\alpha)t - 4.9t^2 \rangle$.



Example 3: Suppose that in two dimensions a projectile is fired from $\langle 2, 5 \rangle$ with a speed of 256 m/s and an angle of elevation of 30° . Where does the projectile hit the ground (round your answer to the hundredth)?

We know from the previous example the position of the particle at time t is

$$\mathbf{r}(t) = \langle 256 \cos(30^\circ)t, 256 \sin(30^\circ)t - 4.9t^2 \rangle + \langle 2, 5 \rangle = \langle 128\sqrt{3}t + 2, 128t - 4.9t^2 + 5 \rangle.$$

The particle hits the ground when the y component of this vector function is 0.

$$\text{So, we solve: } 128t - 4.9t^2 + 5 = 0 \Rightarrow t = \frac{-128 \pm \sqrt{128^2 - 4(-4.9)(5)}}{-9.8} \Rightarrow t \approx 26.16145324.$$

So, the position at which the particle hits the ground is approximately: $\mathbf{r}(26.16145324) \approx \langle 5802.06, 0 \rangle$.

Exploration:

When physicists study the motion of a particle, it is often useful to determine: “the amount of acceleration tangent to the motion” and “the amount of acceleration normal to the motion”. This is similar to when driving in a car.

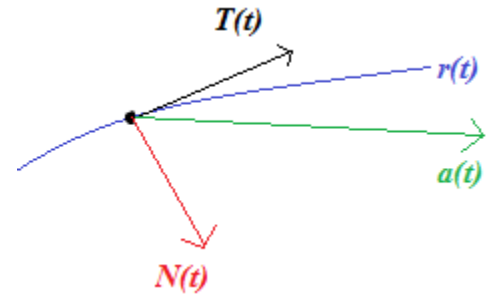
In order to make these two ideas precise, suppose that a particle traverses a smooth curve, C , given by $\mathbf{r}(t)$.

As always, let $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ be the unit tangent vector to C and $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$ the unit normal vector to C .

We want to know the component projection of $\mathbf{a}(t)$ onto $\mathbf{T}(t)$ and the component projection of $\mathbf{a}(t)$ onto $\mathbf{N}(t)$. A picture of the situation is shown at the right.

So, the tangential component of acceleration is $a_T = \text{comp}_{\mathbf{T}}(\mathbf{a})$ and the normal component of acceleration is $a_N = \text{comp}_{\mathbf{N}}(\mathbf{a})$.

In order to give nice formulas for these components of acceleration, we do the following simple calculations.



(Note: throughout the calculation $v(t)$ represents the speed of the particle at time t and $\kappa(t)$ represents the curvature of C at t):

$$\begin{aligned} \mathbf{a}(t) &= \frac{d}{dt}[\mathbf{v}(t)] = \frac{d}{dt}[\mathbf{r}'(t)] = \frac{d}{dt}\left[\mathbf{r}'(t) \frac{\|\mathbf{r}'(t)\|}{\|\mathbf{r}'(t)\|}\right] = \frac{d}{dt}\left[\|\mathbf{r}'(t)\| \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}\right] = \frac{d}{dt}[v(t)\mathbf{T}(t)] \\ &= v'(t)\mathbf{T}(t) + v(t)\mathbf{T}'(t) = v'(t)\mathbf{T}(t) + v(t)\mathbf{T}'(t) \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{T}'(t)\|} = v'(t)\mathbf{T}(t) + v(t)\|\mathbf{T}'(t)\| \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} \\ &= v'(t)\mathbf{T}(t) + v(t)\|\mathbf{T}'(t)\|\mathbf{N}(t) = v'(t)\mathbf{T}(t) + v(t)\|\mathbf{T}'(t)\| \frac{\|\mathbf{r}'(t)\|}{\|\mathbf{r}'(t)\|} \mathbf{N}(t) \\ &= v'(t)\mathbf{T}(t) + v(t)\|\mathbf{r}'(t)\| \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} \mathbf{N}(t) = v'(t)\mathbf{T}(t) + [v(t)]^2 \kappa(t) \mathbf{N}(t) \end{aligned}$$

Amazingly we see that the acceleration of the particle at time t can always be written as a linear combination of the unit tangent vector and the normal tangent vector. Moreover, we can now give nice formulas for the tangential and normal components of acceleration. Specifically,

$$\begin{aligned} a_T = \text{comp}_{\mathbf{T}}(\mathbf{a}) &= \frac{\mathbf{a}(t) \cdot \mathbf{T}(t)}{\|\mathbf{T}(t)\|} = \mathbf{a}(t) \cdot \mathbf{T}(t) = \left[v'(t)\mathbf{T}(t) + [v(t)]^2 \kappa(t)\mathbf{N}(t) \right] \cdot \mathbf{T}(t) \\ &= v'(t)[\mathbf{T}(t) \cdot \mathbf{T}(t)] + [v(t)]^2 \kappa(t)[\mathbf{N}(t) \cdot \mathbf{T}(t)] = v'(t)\|\mathbf{T}(t)\|^2 = v'(t) \end{aligned}$$

and

$$\begin{aligned} a_N = \text{comp}_{\mathbf{N}}(\mathbf{a}) &= \frac{\mathbf{a}(t) \cdot \mathbf{N}(t)}{\|\mathbf{N}(t)\|} = \mathbf{a}(t) \cdot \mathbf{N}(t) = \left[v'(t)\mathbf{T}(t) + [v(t)]^2 \kappa(t)\mathbf{N}(t) \right] \cdot \mathbf{N}(t) \\ &= v'(t)[\mathbf{T}(t) \cdot \mathbf{N}(t)] + [v(t)]^2 \kappa(t)[\mathbf{N}(t) \cdot \mathbf{N}(t)] = [v(t)]^2 \kappa(t)\|\mathbf{N}(t)\|^2 = [v(t)]^2 \kappa(t). \end{aligned}$$

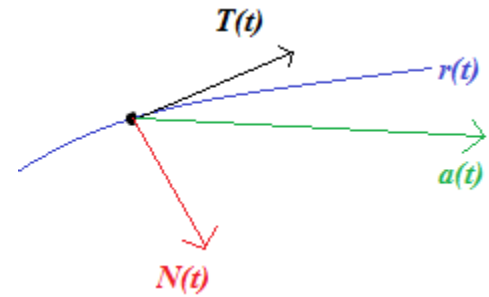
So, in summary, the formulas for the tangential and normal components of acceleration at time t are:

$$a_T(t) = v'(t) \text{ and } a_N(t) = [v(t)]^2 \kappa(t).$$

This of course implies that $\mathbf{a}(t) = a_T \mathbf{T} + a_N \mathbf{N}$.

Note: What we have done in the above exploration is quite profound. We see from our derivations that no matter how an object moves through space, its acceleration always lies in the plane determined by the vectors $\mathbf{T}$ and $\mathbf{N}$. Physicists refer to this plane as the osculating plane.

Note: Understanding the scalar components of acceleration allow us to explain the effect you experience when a car speeds up quickly or turns. Imagine you are in a car traveling along the curve $\mathbf{r}(t)$. Speeding up will



produce large values for a_T and therefore create a large tangential scalar component of acceleration.

When rounding a turn, the curvature and velocity will play an important role in the normal forces on the car. Therefore, taking sharp turns at high speeds will correspond to large normal forces. Notice that both of these descriptions can seem to be opposite to what you feel when you are riding in a car, but that is because you are feeling the forces on you and these forces are in the opposite direction of what the car is experiencing. When the car speeds up (increased tangential component) you feel pushed back and when the car rounds a sharp corner quickly (increased normal component) you feel pushed to the outside of the turn.

Example 4: Suppose that the motion of a particle is determined by: $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$. Given that

$\kappa(t) = \frac{1}{2}$, find the tangential and normal components of the acceleration of the particle at time t .

We calculate that: $\mathbf{v}(t) = \langle -\sin(t), \cos(t), 1 \rangle$.

Then, $v(t) = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$ and $v'(t) = 0$.

So, $a_T(t) = 0$ and $a_N(t) = [\sqrt{2}]^2 \frac{1}{2} = 1$.